

Measuring Neural-Network Robustness Under Adversarial Perturbation

Experimental Evaluation of Model Behavior Under Gradient-Based Attacks

github.com/poojakira/adversarial-ml-lab - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

109

Tests passed

32.16%

Statement coverage

1,117,354

SmallCNN parameters

71.82%

Clean CIFAR-10 accuracy

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface

Data, models, requests, policies, or artifacts entering the tested path.

Control surface

Repository-specific validation, analysis, and fail-closed checks.

Evidence surface

Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01

Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget.

02

Measure clean accuracy on the full CIFAR-10 test set.

03

Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples.

04

Measure robust accuracy on the committed attack subset.

05

Emit structured evidence and CI robustness-gate logic.

Research Contribution

A reproducible robustness-evaluation harness for FGSM, PGD, and C&W that shows a measured 71.82% clean CIFAR-10 SmallCNN collapsing to 0.00% robust accuracy under PGD-20 at epsilon 8/255, while explicitly separating measured results from literature projections.

Verified Evidence

EV-01	109 tests passed.	EV-02	32.16% statement coverage; CI gate is 15%.
EV-03	Lint/format, security audit, and CodeQL jobs succeeded.	EV-04	The current CI robustness benchmark job is skipped on ordinary pushes, so the benchmark values below come from the committed real artifact rather than this CI run.
EV-05	FGSM robust accuracy @ 8/255: 3.32%	EV-06	PGD-20 robust accuracy @ 8/255: 0.00%

Method and Control Path

- 01 Train/evaluate a compact CIFAR-10 classifier under a fixed seed and CPU budget.
- 02 Measure clean accuracy on the full CIFAR-10 test set.
- 03 Generate epsilon-bounded FGSM and PGD adversarial samples plus C&W L2 samples.
- 04 Measure robust accuracy on the committed attack subset.
- 05 Emit structured evidence and CI robustness-gate logic.

Limitations and Residual Risk

- This repository measures vulnerability; it does not make a model robust.
- Robust-accuracy attacks use a 1,024-sample subset, not the full 10,000 test set.
- No AutoAttack, black-box transfer, or certified-robustness result is established.
- results/cifar10_resnet18_benchmark.json is explicitly literature-projected/synthetic and must not be presented as measured.

Reproducibility

Expected current CI evidence: 109 passed, 32.16% coverage.

Evidence Provenance

- Claim discipline**
Metrics come from the repository's verified metrics and research brief.
- Scope discipline**
Benchmarks are reported only at their documented fixture, dataset, or test scope.
- Reproduction**
PDF and PNG are generated locally from committed evidence without paid services.